context_compactor.sh — Companion Guide

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Overview

§11.4.141 M7 implementation. Prunes stale tool-results and old thinking from long sessions. NEVER touches the governance prefix.

Functions

Function	Purpose	Example
`cc_should_compact`	Check if context exceeds threshold	`cc_should_compact 50`
`cc_prune_stale_results`	Mark old results as prunable	`cc_prune_stale_results 20`
`cc_summarize_old_thinking`	Compress old thinking blocks	`cc_summarize_old_thinking`
`cc_preserve_governance`	Documented invariant (no-op)	`cc_preserve_governance`
`cc_get_context_stats`	Return context statistics	`cc_get_context_stats`

Usage

```
source constitution/scripts/token_efficiency/context_compactor.sh

# Check if compaction needed (after 50 turns)
if cc_should_compact 50; then
    cc_prune_stale_results 20
    cc_summarize_old_thinking
fi

# Governance preservation is automatic (invariant)
```

Invariant

The governance prefix (CLAUDE.md, Constitution.md) is NEVER touched by compaction. It is the cache breakpoint and must remain byte-stable for prompt caching.

Cross-references

- Constitution §11.4.141 M7
- `token_efficiency_lib.sh` — for status and caching
- `scoped_read.sh` — for targeted reads